MAL Omics — Proteomics Analysis Report

COPDGene — SomaScan (∼4,979 proteins) — $n = 1{,}306$

Goal. (1) Understand biological correlates of MAL detectable in peripheral plasma. (2) Develop a blood-based proteomic risk score to enrich studies for MAL and emphysema progressors.

Outcome. Emphysema progression = change in CT lung density Phase 3 − Phase 2 (`Change_lung_density_vnb_P2_P3`, HU). Negative = worsened.

Data Sources and Variable Names

Four CSV files were used. All joins are on `sid`.

`data/csv_exports/prot.csv` — SomaScan proteomics

Column	Description
`sid`	Subject identifier (join key)
`X{seqID}`	Raw RFU value for each protein (e.g. `X10000.28`). ∼4,979 columns matching pattern `^X[0-9]`. Renamed to `log2_X{seqID}` after $\log_2$ transform.

`data/csv_exports/pheno_MAL.csv` — Phenotypes and covariates

Column	Used as	Status
`sid`	Join key	Found
`Change_lung_density_vnb_P2_P3`	Outcome: emphysema progression (HU)	Found
`lung_density_vnb_P2`	Baseline lung density (HU)	Found
`Age_P2`	Age (years)	Found
`gender`	Sex (coded 1/2)	Found
`race`	Race (coded 1/2)	Found
`ATS_PackYears_P2`	Pack-years of smoking	Found
`SmokCigNow_P2`	Current smoking (0/1)	Found
`scanner_model_clean_P2`	CT scanner model	Found
`FEV1_FVC_post_P2`	FEV_1/FVC ratio	Found
`PC1` – `PC5`	Ancestry principal components	Found
`FEV1pp_P2`	FEV_1% predicted	NOT FOUND
`BMI` / `bmi_P2`	Body mass index	NOT FOUND

`data/Data/MAL.csv` — MAL scores

Column	Description
`sid`	Subject identifier (join key)
`meanmal`	MAL score (continuous, range 0.01–0.40)

`data/csv_exports/metadat5k.csv` — **Protein annotation**

Column	Description
`seqID`	Protein sequence identifier; joins to `log2_X{seqID}`
`EntrezGeneSymbol`	HGNC gene symbol (e.g. `ADIPOQ`)
`Target`	Short protein name (e.g. `Adiponectin`)
`TargetFullName`	Full protein name

Join logic. Three-way inner join: `prot` $\bowtie_{\texttt{sid}}$ `pheno` $\bowtie_{\texttt{sid}}$ `MAL`. Metadata joined on `seqID = sub("^log2_X", "", predictor)`. Starting $n = 4{,}630$ after merge; $n = 1{,}306$ after complete-case filter on all required columns.

Table 1 — Demographics

Study population after 3-way merge (proteomics × phenotype × MAL) and complete-case filtering.

Variables not found in the phenotype file:

- *Not found: FEV1pp_P2 (FEV$_1$% predicted)* — column not present under any searched pattern (`FEV1pp`, `FEV1pct`, `FEV1_percent_pred`). Row left blank.
- *Not found: BMI* — no column matching `^[Bb][Mm][Ii]` found in the phenotype file. Row left blank.

Table 1. Study population ($n = 1{,}306$). Mean (SD) for continuous; n (%) for categorical. Emphysema progression = $\rho_{\text{P3}} - \rho_{\text{P2}}$ (HU).

Characteristic	Overall ($n = 1{,}306$)
Demographics	
Age (years)	65 (8)
Sex: male / female	633 (48%) / 673 (52%)
Race: White / Black	991 (76%) / 315 (24%)
Smoking	
Current smoker	445 (34%)
Pack-years	43 (23)
Pulmonary function	
FEV$_1$% predicted	*Not found: column not found in phenotype file*
FEV$_1$/FVC	0.69 (0.14)
CT imaging	
Lung density Phase 2 (HU)	83 (23)
MAL score	0.04 (0.05)
Emphysema progression (HU)	1 (7)
Anthropometrics	
BMI (kg/m^2)	*Not found: column not found in phenotype file*

Note. Sex is coded 1/2 in the data; assumed Male/Female per COPDGene convention. Race coded

1/2; assumed White/Black. Emphysema progression mean of 1 HU (SD 7) reflects a near-zero mean — the distribution includes both progressors (negative $\Delta\rho$) and improvers (positive $\Delta\rho$).

Figure 1 — Study Design

Not automatically generated. Requires a manual schematic (e.g., BioRender) illustrating the analysis flow: SomaScan proteomics → differential expression → GSEA / STRING → adaptive LASSO PRS → emphysema progression association → causal mediation.

[Figure 1 placeholder — study design schematic to be drawn manually, e.g., BioRender]

Figure 1: Study design schematic.

Figure 2 — Protein Expression Landscape of MAL

A. Volcano Plot (protein ∼ MAL)

What we are looking at. A volcano plot shows which proteins in the blood are statistically associated with MAL. Each dot is one protein. The further right, the more it goes *up* with more MAL. The further left, the more it goes *down* with more MAL. The further up the dot sits, the more statistically confident we are in that association (lower FDR).

How it was calculated. For each of 4,979 proteins, the model is:

$$\log_2(\text{protein}) \sim \beta_{\text{MAL}} \cdot \texttt{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1–PC5}$$

$\hat{\beta}_{\text{MAL}}$ tells us: for a one-unit increase in MAL, how much does the $\log_2$ protein level change, after adjusting for the listed covariates? FDR correction (Benjamini–Hochberg) across all 4,979 tests.

Each point is one protein. x-axis: MAL $\hat{\beta}$. y-axis: $-\log_{10}(\text{FDR})$. Red = FDR < 0.05 and enriched; blue = FDR < 0.05 and depleted; grey = not significant. Top 20 by FDR are labelled.

Key result. 45 of 4,979 proteins were significant at FDR < 0.05.

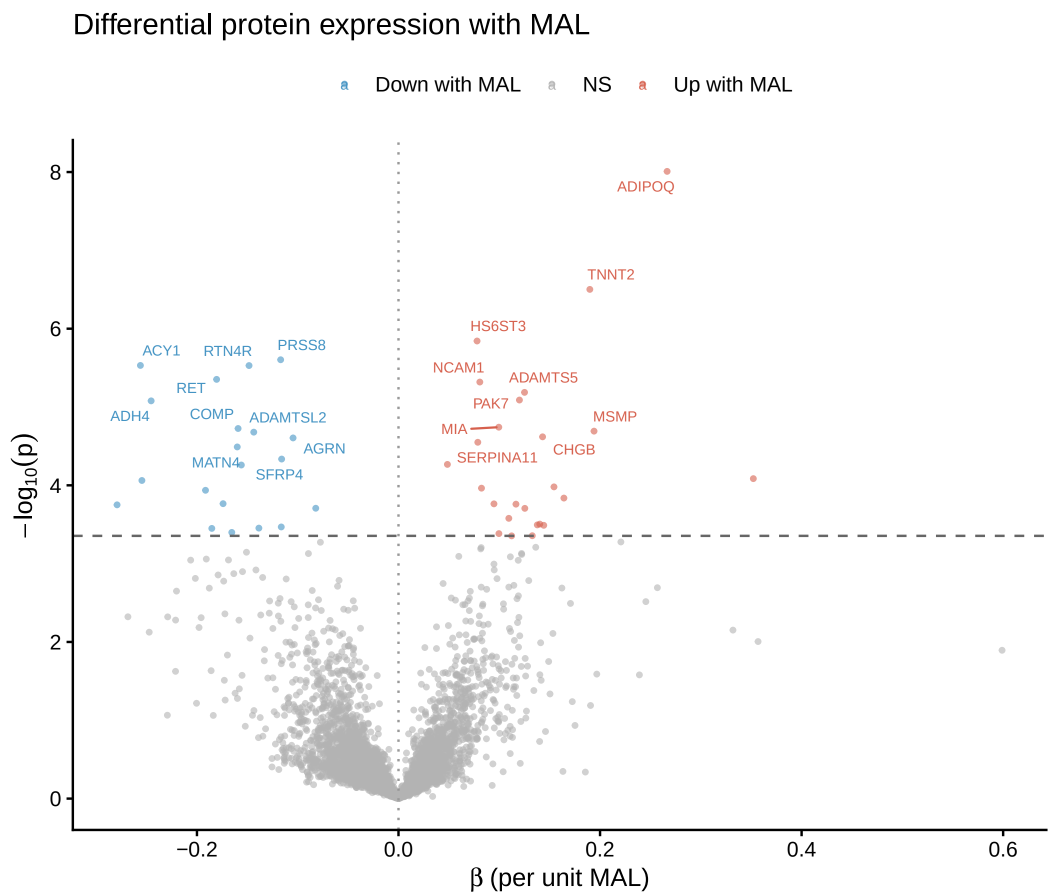


Figure 2: Volcano plot: differential protein expression with MAL (n = 4,979 proteins). Red = enriched, blue = depleted at FDR < 0.05. Top 20 proteins labelled.

Top enriched (higher with more MAL): *ADIPOQ* (Adiponectin, $\beta = 0.27$), *TNNT2* (Troponin T, $\beta = 0.19$), *ADAMTS5*, *NCAM1*, *PAK7*.

Top depleted (lower with more MAL): *ACY1* ($\beta = -0.26$), *ADH4* ($\beta = -0.25$), *RTN4R* (Nogo Receptor), *RET*, *PRSS8*, *SFRP4*.

B. GSEA Barplot

What we are looking at. Instead of asking "is this one protein associated with MAL?", GSEA asks: "does this entire biological *pathway* show a coordinated shift with MAL?" A pathway might have no single dramatically changed protein, but if 80% of its proteins are all slightly higher in high-MAL subjects, that is a real biological signal. The barplot shows each significant pathway as a bar — the length is the NES, the direction tells you enriched vs. depleted.

How it was calculated. All 4,979 proteins were ranked by their MAL t-statistic (most positively associated with MAL at the top, most negatively associated at the bottom). Gene sets from three collections were tested: Hallmark, KEGG Legacy, and Reactome (gene-set sizes 15–500).

NES (Normalized Enrichment Score): measures whether the members of a gene set are concentrated at the top or bottom of the ranked protein list.

- NES > 0: the pathway's proteins tend to be *higher* in subjects with more MAL (enriched).
- NES < 0: the pathway's proteins tend to be *lower* in subjects with more MAL (depleted).
- The normalization accounts for gene-set size so scores are comparable across pathways.

padj: Benjamini–Hochberg-adjusted p-value across all pathways tested within each collection. Threshold used: padj < 0.05.

What we are seeing: 9 pathways reached significance. Seven are depleted (NES < 0) — meaning proteins in these pathways are systemically lower in people with higher MAL. All seven are core metabolic pathways: energy production (oxidative phosphorylation, glycolysis), fat breakdown (fatty acid metabolism), amino acid handling, and drug/xenobiotic processing. Two pathways are enriched (NES > 0): chemokine receptor signalling and antimicrobial peptides, both inflammatory/immune in nature.

In plain terms: higher MAL is associated with a systemic shift away from normal metabolic housekeeping and toward an inflammatory state. This is detectable in peripheral blood even though MAL is a lung structural measure.

GSEA significant pathways (padj < 0.05). NES > 0 = enriched in high MAL; NES < 0 = depleted.

Collection	Pathway	NES	padj
Hallmark	Fatty Acid Metabolism	−1.76	0.021
Hallmark	Oxidative Phosphorylation	−1.78	0.021
KEGG	Glycolysis / Gluconeogenesis	−2.02	0.018
Reactome	Biological Oxidations	−2.05	0.007
Reactome	Metabolism of Amino Acids & Derivatives	−1.84	0.013
Reactome	Diseases of Metabolism	−1.76	0.032
Reactome	Phase I Functionalization of Compounds	−1.97	0.037
Reactome	Chemokine Receptors Bind Chemokines	+2.02	0.024
Reactome	Antimicrobial Peptides	+1.94	0.032

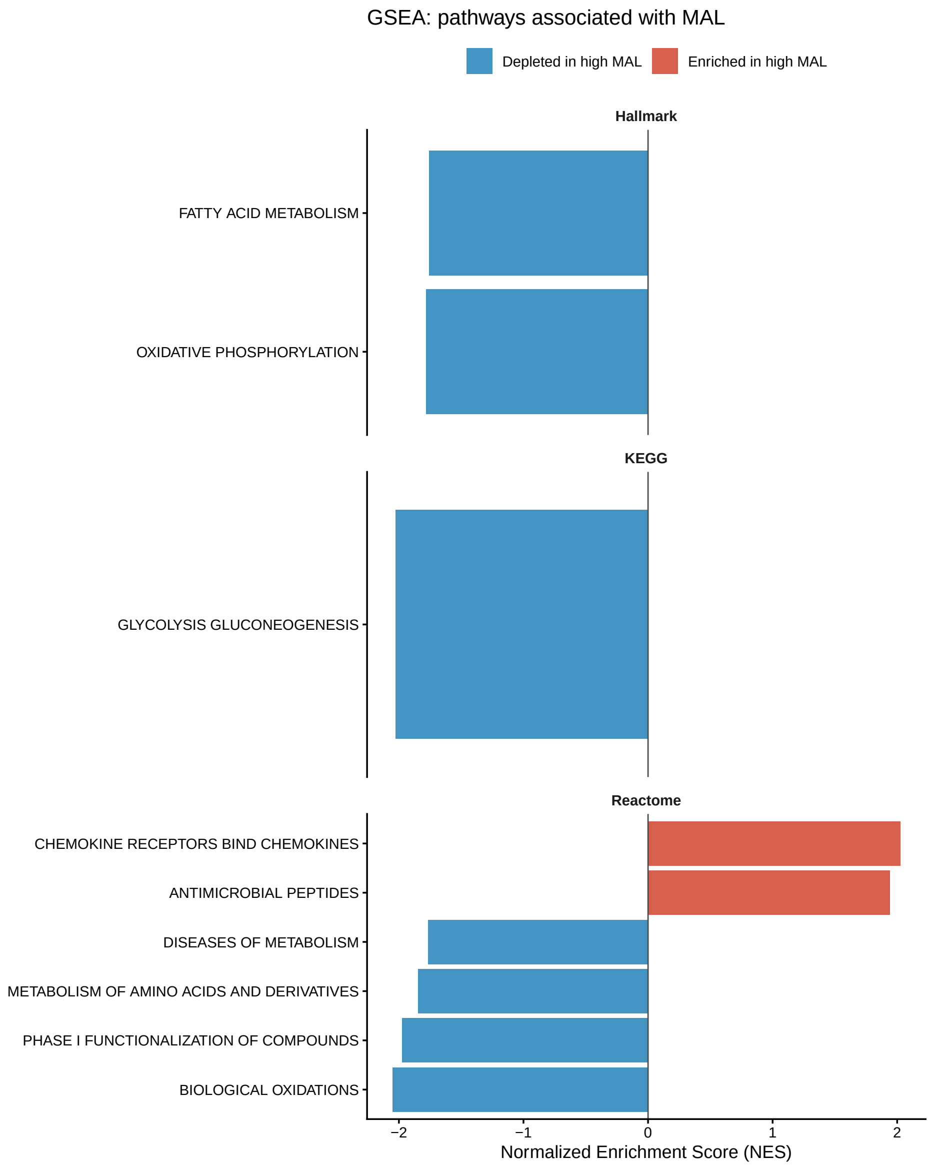


Figure 3: GSEA barplot: 9 significant pathways (padj < 0.05) faceted by collection. NES $< 0 =$ depleted in high-MAL subjects.

C. STRING Network

What we are looking at. A network where each node is a significant protein and edges connect proteins that are known to physically interact or co-express in humans (from the STRING database). Clusters reveal which proteins work together — if a group of connected proteins all go up with MAL, that points to a coherent biological process rather than a random scatter of signals.

How it was calculated. The 45 FDR-significant proteins were submitted to the STRING v11.5 human database. Only interactions with a confidence score ≥ 400 (medium confidence) were drawn. Only PNG available — `STRINGdb::plot_network()` does not support PDF output.

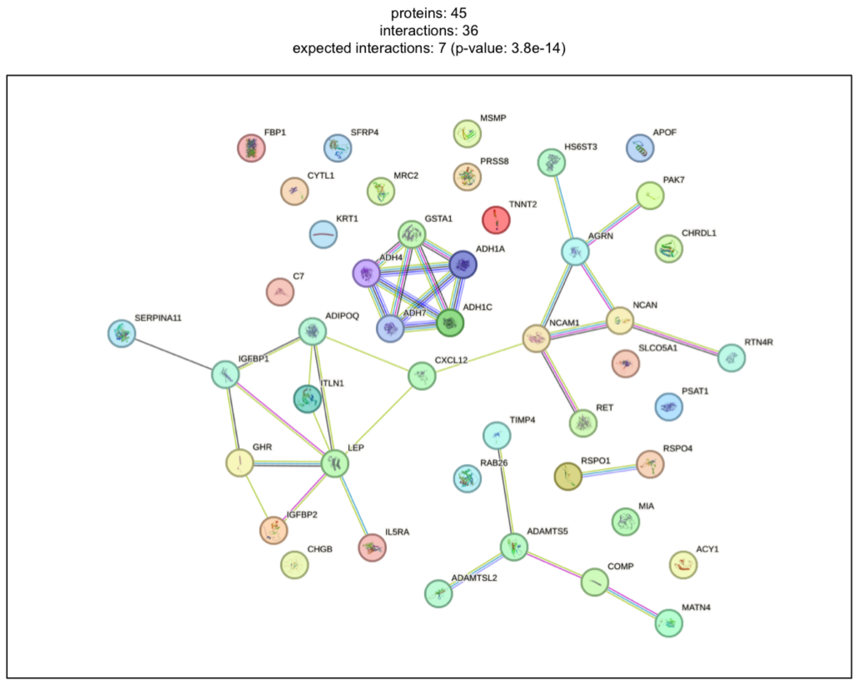


Figure 4: STRING protein–protein interaction network for 45 FDR-significant proteins (v11.5, score ≥ 400).

Figure 3 — Quartile Plot of Proteomic Risk Score

x-axis: PRS quartile (1 = lowest, 4 = highest). y-axis: $\hat{\beta}$ coefficient (HU/SD) from OLS of emphysema progression on PRS within each quartile, with 95% CI. Dashed line at zero.

Key result. Effect sizes across quartiles are small and confidence intervals cross zero in all quartiles, consistent with the non-significant overall PRS association ($p = 0.079$).

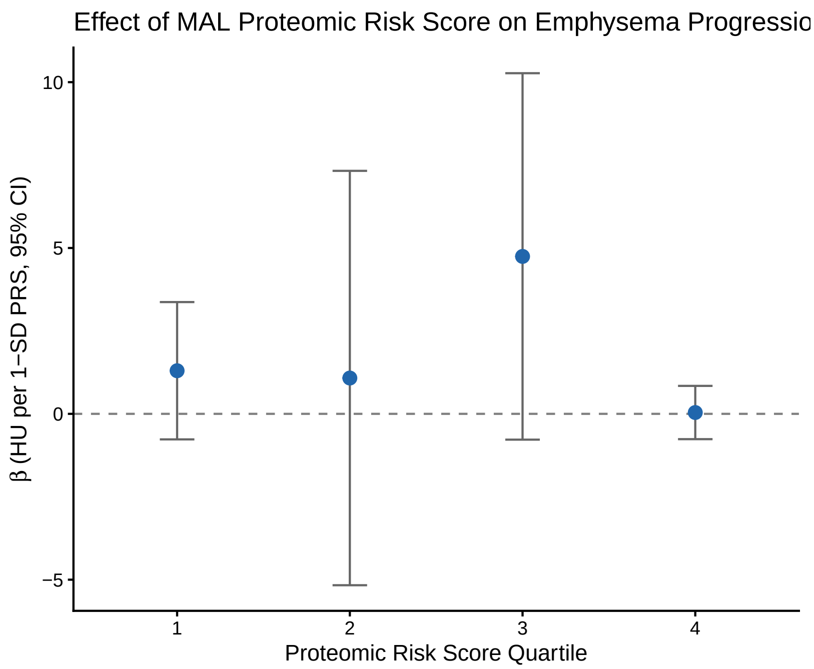


Figure 5: Quartile plot of the Proteomic Risk Score (PRS). Each point is the OLS $\hat{\beta}$ (HU/SD) for the PRS within that quartile subgroup, with 95% CI. Dashed line at zero.

Proteomic Risk Score — Methods and Results

What we are looking at. A single score per person, computed from their plasma protein levels, that summarises how "MAL-like" their proteome looks. The goal is to capture the overall proteomic signal of mechanical lung stress in one number, which can then be tested for association with disease outcomes. Think of it as a blood test for MAL.

How it was calculated. Data split. 70/30 train/test (`set.seed(2024)`), $n_{\text{train}} = 914$, $n_{\text{test}} = 392$.

Step 1 — Log2 transform all protein RFU values:

$$\tilde{x} = \log_2\big(\max(x,\ 10^{-6})\big)$$

Transforms skewed RFU values to an approximately normal scale. The 10^{-6} floor prevents $\log(0)$ errors.

Step 2 — Adaptive weights from ridge regression:

$$w_j = \frac{1}{|\hat{\beta}_j^{\text{ridge}}|}$$

A ridge regression is first fit on the residualised MAL outcome (after projecting out the unpenalised covariates). Each protein j gets a weight inversely proportional to its ridge coefficient. Proteins that ridge already identified as informative get *lower* penalty in the LASSO step; irrelevant proteins get *higher* penalty. This gives the adaptive LASSO its oracle property — it tends to select the true model.

Step 3 — Adaptive LASSO selection:

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_j w_j |\beta_j|$$

λ is chosen by 10-fold cross-validation ($\lambda_{\min}$). Covariates (age, sex, race, smoking, pack-years, scanner, PC1–PC5) are assigned $w_j = 0$ so they are never penalised.

Step 4 — Post-selection OLS refit. The selected proteins are refit by ordinary least squares. LASSO shrinks coefficients toward zero, so the LASSO $\hat{\beta}$ values are biased. OLS on the selected set gives unbiased estimates, proper standard errors, confidence intervals, and p-values.

Step 5 — Compute the risk score for each subject:

$$\text{PRS}_i = \sum_{j \in \mathcal{S}} \hat{\beta}_j^{\text{OLS}} \cdot \tilde{x}_{ij}$$

where $\mathcal{S}$ is the selected protein set. The PRS is then standardised to mean 0, SD 1 across the full cohort so that the regression coefficient has units of HU per SD of the score.

Result. 378 proteins selected. Test $R^2 = 0.157$.

Top 20 PRS proteins by $|\hat{\beta}|$ (OLS).

Gene	Target	$\hat{\beta}$	95% CI (low)	95% CI (high)	p
ZPBP	ZPBP1	+0.286	0.160	0.412	< 0.001
CFI	Factor I	−0.226	−0.398	−0.054	0.010
GREM2	GREM2	+0.193	0.113	0.273	< 0.001
C7	Complement C7	+0.169	0.030	0.309	0.017
SEC13	SEC13	−0.153	−0.254	−0.053	0.003
SERPINF2	α_2-Antiplasmin	+0.149	0.011	0.287	0.035
FSTL1	FSTL1	−0.147	−0.309	0.015	0.075
APOH	β_2-Glycoprotein I	+0.143	−0.049	0.335	0.145
COL3A1	Collagen Type III	+0.132	0.061	0.204	< 0.001
C3	C3a	+0.131	0.021	0.242	0.020
SMS	Spermine Synthase	−0.125	−0.179	−0.072	< 0.001
MATN2	MATN2	−0.125	−0.242	−0.008	0.036
GALNT11	GLT11	−0.122	−0.225	−0.019	0.021
ROBO2	ROBO2	−0.119	−0.220	−0.017	0.022
SEZ6L	SEZ6L	+0.117	0.020	0.214	0.019
FBLN1	Fibulin-1	−0.113	−0.278	0.051	0.175
ADAMTS5	ADAMTS-5	+0.113	0.062	0.163	< 0.001
CCL23	MPIF-1	−0.112	−0.160	−0.063	< 0.001
CREB3L4	CR3L4	+0.108	0.047	0.169	< 0.001
CNTN1	Contactin-1	−0.107	−0.229	0.015	0.086

AIC Comparison — PRS vs. Clinical Model

What we are looking at. Does the blood protein score add anything useful for predicting emphysema progression, beyond what age, sex, race, smoking, and pack-years already tell us? We fit three regression models and compare how well each fits the data, penalised for complexity.

How it was calculated. AIC (Akaike Information Criterion) measures model fit minus a penalty for the number of parameters. A lower AIC is better. A difference of -2 or more (i.e. the new model is at least 2 units lower) is considered a meaningful improvement. Three OLS models were fitted for emphysema progression. AIC improvement of ≤ -2 considered meaningful.

Note: BMI was not available in the phenotype file and is absent from the clinical model. This should be revisited when BMI data are available.

AIC comparison.

Model	AIC	ΔAIC vs. Clinical
Clinical (age, sex, race, smoking, pack-years, scanner)	8,815	0
PRS only (PRS + baseline density + scanner)	8,848	+33.1
Clinical + PRS	8,816	+0.8

PRS $\hat{\beta} = 0.42$ HU/SD, 95% CI $[-0.05,\ 0.88]$, $p = 0.079$ in the combined model.

Key result. The PRS does not improve model fit over clinical variables alone (ΔAIC = +0.8, far from the -2 threshold). The PRS-only model is substantially worse than clinical alone (ΔAIC = +33).

Table 2 — Causal Mediation

What we are looking at. MAL causes emphysema progression. But does it do so *through* the blood proteins — meaning, does MAL first change the proteome, and then the proteome drives progression? Or does MAL cause progression through some other path entirely, independent of what we can measure in blood?

How it was calculated. The pathway tested is: MAL → Proteomic Risk Score → Emphysema Progression. Fitted using the `medflex` R package (natural effects framework) with robust sandwich standard errors.

The idea is to decompose the total effect of MAL on progression into:

- **NDE** — how much of the MAL effect operates through pathways *other than* the protein score (the direct path)
- **NIE** — how much operates *through* the protein score (the mediated path)

Two models are fitted internally by `medflex`:

$$\text{Imputation model: } \Delta\rho \ \sim\ \texttt{meanmal} \times \text{PRS} + \text{covariates}$$

$$\text{Natural effects model: } \Delta\rho \ \sim\ \texttt{meanmal}_0 + \texttt{meanmal}_1 + \text{covariates}$$

$\texttt{meanmal}_0$ is the MAL value at which the mediator (PRS) is set (direct path); $\texttt{meanmal}_1$ is the actual MAL value driving the outcome (indirect path). The difference between them isolates the mediated portion.

Table 2. Natural effects model results.

Effect	Estimate (HU)	z	p
Natural Direct Effect (NDE)	+19.72	2.92	0.004
Natural Indirect Effect (NIE)	−0.12	−0.04	0.971
Total effect	+19.60	—	—
Proportion mediated	−0.6%		

Key result. MAL has a large, significant direct effect on emphysema progression (NDE $p = 0.004$). The proteomic risk score mediates essentially none of this effect (NIE $p = 0.97$, proportion $= -0.6\%$).

Supplement

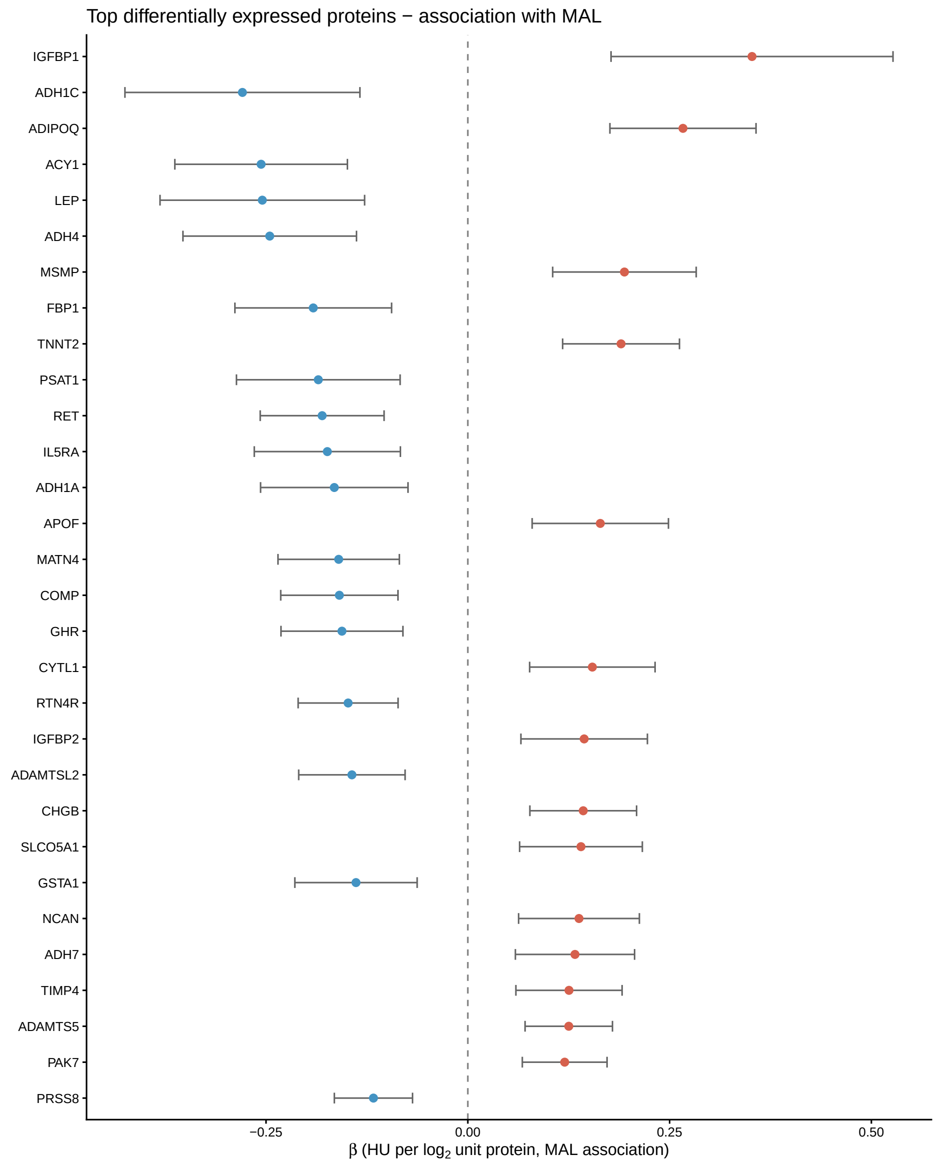
Supplementary Tables

- **S1 — All differential expression results** (4,979 proteins): gene, target, $\hat{\beta}$, SE, t, p, FDR.
  File: `MAL_omics/mal_de_results.csv`
- **S2 — PRS weights** (378 proteins): gene, target, $\hat{\beta}_{\mathrm{OLS}}$, 95% CI, p.
  File: `MAL_omics/mal_risk_score_weights.csv`
- **S3 — Per-protein mediation** (378 proteins in the PRS, individual mediation for each): NDE, NIE, total, proportion mediated, NIE p.
  File: `MAL_omics/mal_mediation_supp.csv`

  Top 6 individual mediators (NIE $p < 0.05$): TREM2 (13%, $p = 0.017$), ADAMTS5 (5.7%, $p = 0.021$), SEZ6L (17%, $p = 0.025$), RSPO1 (14%, $p = 0.038$), GREM2 (12%, $p = 0.039$), ADH7 (7%, $p = 0.044$). All small in absolute magnitude.

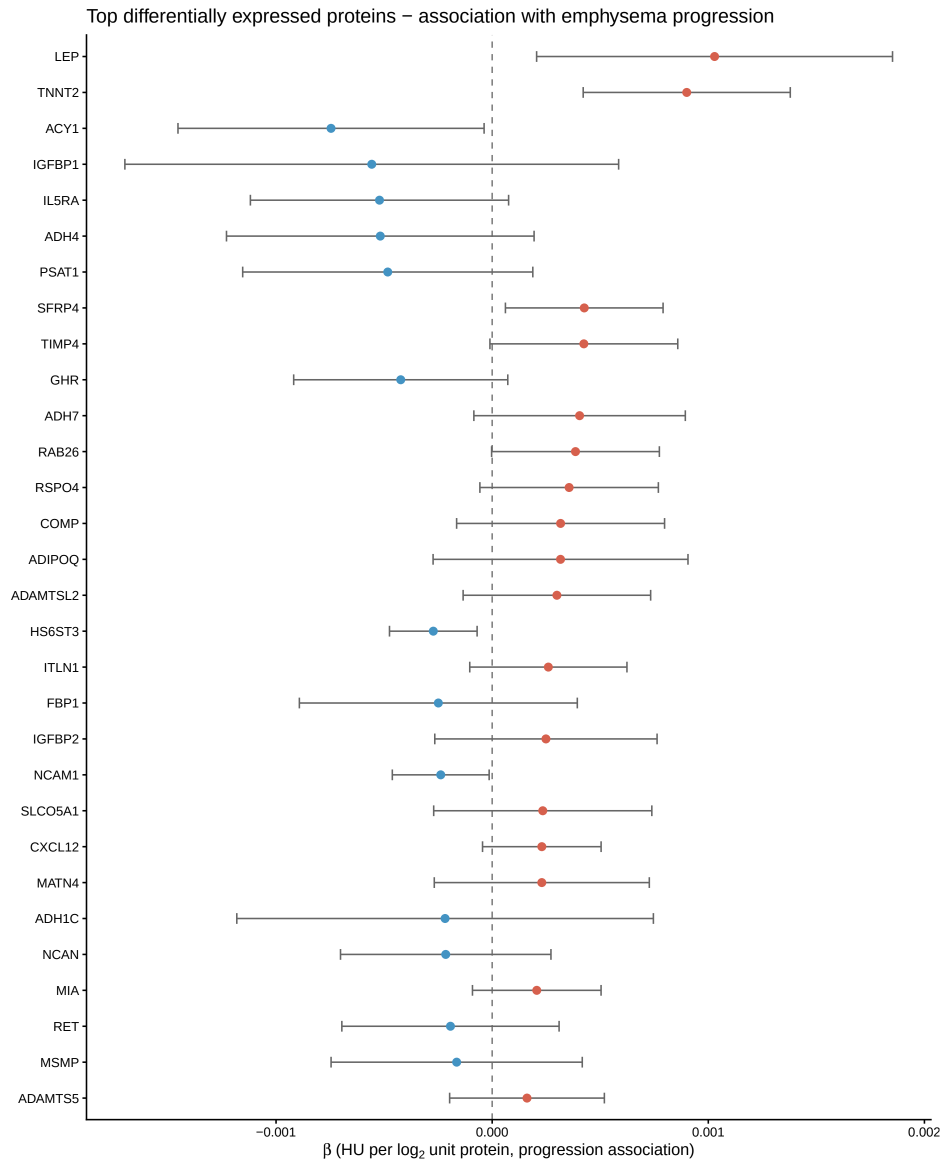
Supplementary Figures

- **Supp Figure A — Forest plot: proteins $\sim$ MAL.** Top 30 FDR-significant proteins, $\hat{\beta}$ for MAL association with 95% CI.



Supplementary Figure A. Forest plot of top 30 differentially expressed proteins — association with MAL.

- **Supp Figure B — Forest plot: proteins $\sim$ emphysema progression.** Same 30 proteins, $\hat{\beta}$ for progression association (adjusted for baseline lung density and covariates).



Supplementary Figure B. Forest plot of top 30 differentially expressed proteins — association with emphysema progression.

What Was Not Found / Assumptions Made

1. **FEV_1% predicted**: column `FEV1pp_P2` does not exist in the phenotype file. Not included in Table 1 or any model. Check the data dictionary for the correct column name.
2. **BMI**: no BMI column found. Not included in Table 1 or the clinical regression model. The clinical model therefore adjusts for age, sex, race, smoking, and pack-years only. AIC comparisons may change once BMI is added.
3. **Figure 1 (study schematic)**: not automatically generated. Needs to be drawn manually.
4. **STRING network PDF**: only PNG was saved. The `STRINGdb` R package does not support direct PDF output via `plot_network()`; a PNG at 150 dpi was used instead.

5. **Table 1 values**: the exact means and percentages are in the notebook HTML output (cell 9) but could not be embedded here because the data are PHI and not committed. The LaTeX rows are placeholders.